

Given:

$$n = 200 \text{ rpm}$$

$$P_A = 10 \text{ kW}$$

$$P_E = 5 \text{ kW}$$

$$P_D = P_A + P_E$$

$$= 10 + 5 = 15 \text{ kW}$$

$$D_A = 200 \text{ mm}$$

$$D_B = 150 \text{ mm}$$

$$D_E = 100 \text{ mm}$$

AISI 1137 CD,  $N=3$ ,  $C_s=0.82$ , Reliability = 99%

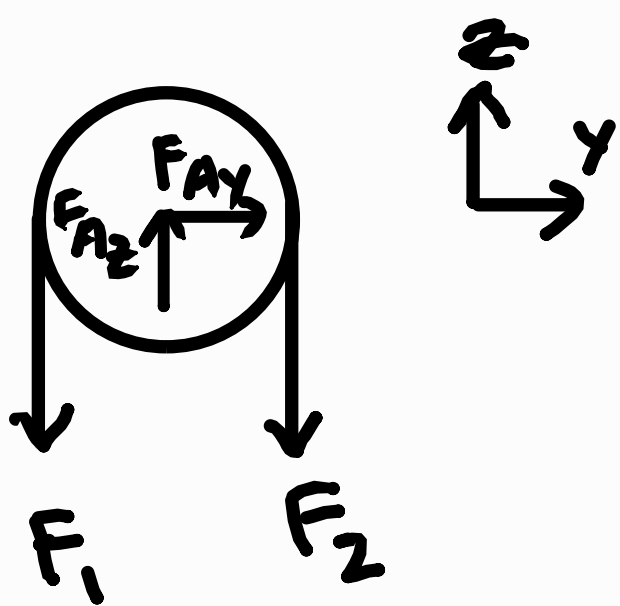
Finding torque:

$$T_A = \frac{P_A}{\omega} = \frac{10 \times 10^3}{\frac{2\pi}{60} \times 200} = 477.46 \text{ Nm} (\downarrow)$$

$$T_E = \frac{P_E}{\omega} = \frac{5 \times 10^3}{\frac{2\pi}{60} \times 200} = 238.73 \text{ Nm} (\downarrow)$$

$$T_D = \frac{P_D}{\omega} = \frac{15 \times 10^3}{\frac{2\pi}{60} \times 200} = 716.20 \text{ Nm} (\uparrow)$$

Finding forces:



$$T_A = (F_1 - F_2) \frac{D}{2}$$

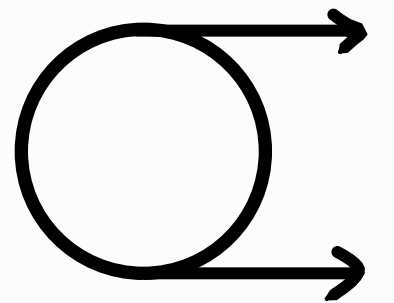
$$F_{net} = \frac{-T_A}{\frac{1}{2}D} = -\frac{477.46}{\frac{1}{2} \times 0.2} = -4774.6 \text{ N}$$

$$F_1 = 5F_2 \quad \therefore F_{net} = 5F_2 - F_2 = 4F_2$$

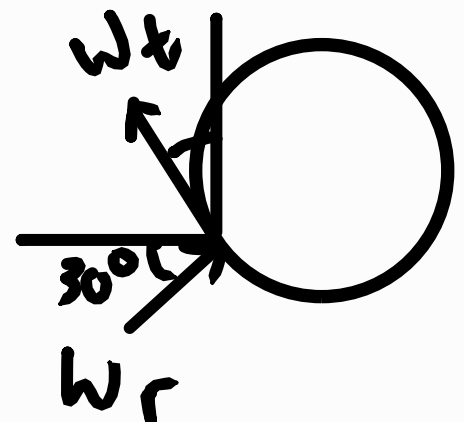
$$F_A = F_1 + F_2 = 5F_2 + F_2 = 6F_2 = \frac{6}{4} F_{net}$$

$$\therefore F_A = \frac{6}{4} F_{net} = \frac{6}{4} (-4774.6) = -7161.9 \text{ N} \downarrow$$

Finding forces:



$$F_2 = 0 \quad F_E = F_1 = \frac{T_E}{\frac{1}{2}} = \frac{238.73}{\frac{1}{2} \times 0.1} = 4774.6 \text{ N} \rightarrow$$



$$W_t = \frac{T_D}{\frac{1}{2}} = \frac{716.20}{\frac{1}{2} \times 0.15} = 9549.33 \text{ N}$$

$$W_r = W_t \tan \phi = 9549.33 \tan 20^\circ = 3475.66 \text{ N}$$

$$F_{D_z} = W_t \cos 30^\circ + W_r \sin 30^\circ$$

$$= 9549.33 \cos 30^\circ + 3475.66 \sin 30^\circ$$

$$= 10067.79 \text{ N} \uparrow$$

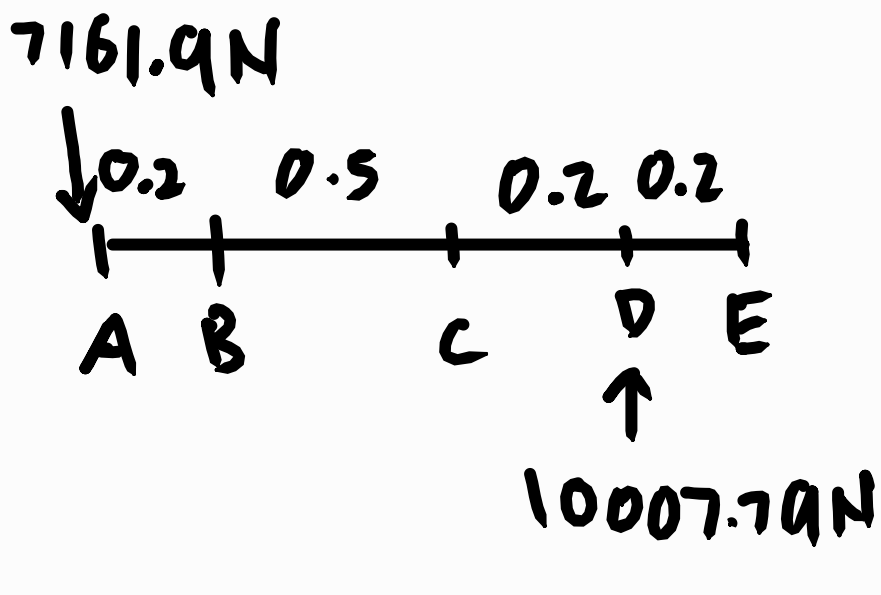
$$F_{D_y} = W_r \cos 30^\circ - W_t \sin 30^\circ$$

$$= 3475.66 \cos 30^\circ - 9549.33 \sin 30^\circ$$

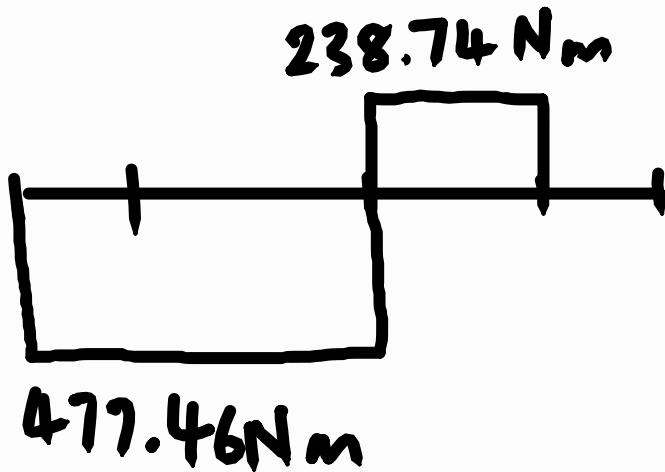
$$= -1764.66 \text{ N} \leftarrow$$

# Bearing forces:

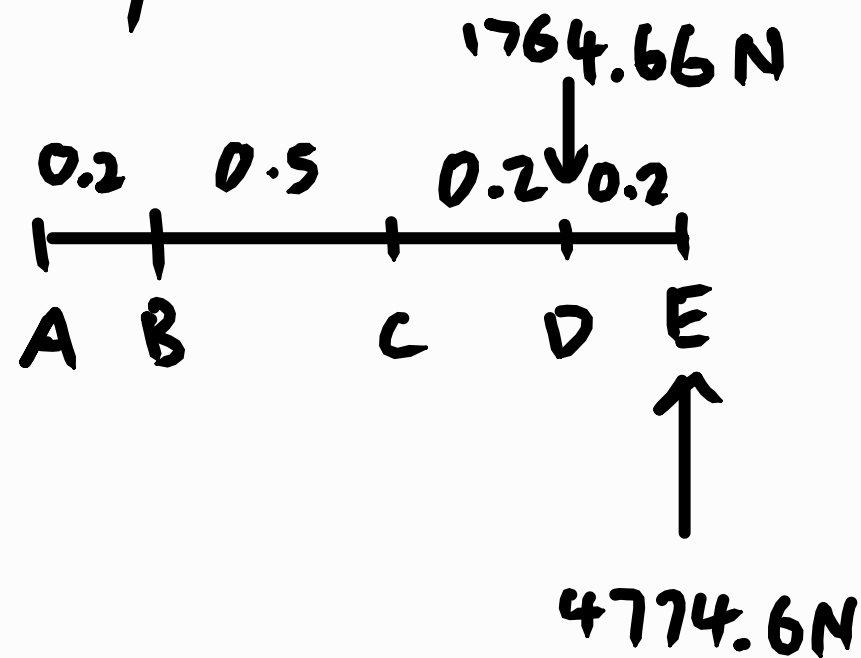
z-direction



Torque diagram:



y-direction



$$\sum M_C = 0$$

$$7161.9(0.7) + F_{B_z}(0.5) + 0.2(10007.79) = 0$$

$$F_{B_z} = 14029.78 \text{ N } \uparrow$$

$$\sum F_z = 0$$

$$-7161.9 + 14029.78 + F_{C_z} + 10007.79 = 0$$

$$F_{C_z} = -16875.67 \text{ N } \downarrow$$

$$\sum M_C = 0$$

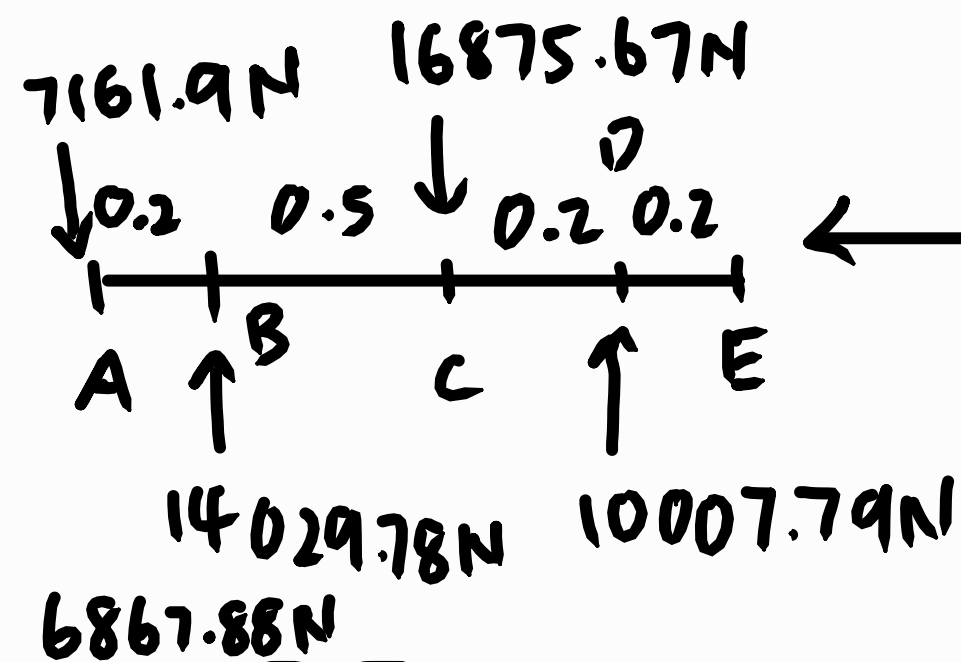
$$1764.66 \times 0.2 = 4774.6 \times 0.4 + 0.5 F_B$$

$$F_{B_y} = 3113.82 \text{ N } \uparrow$$

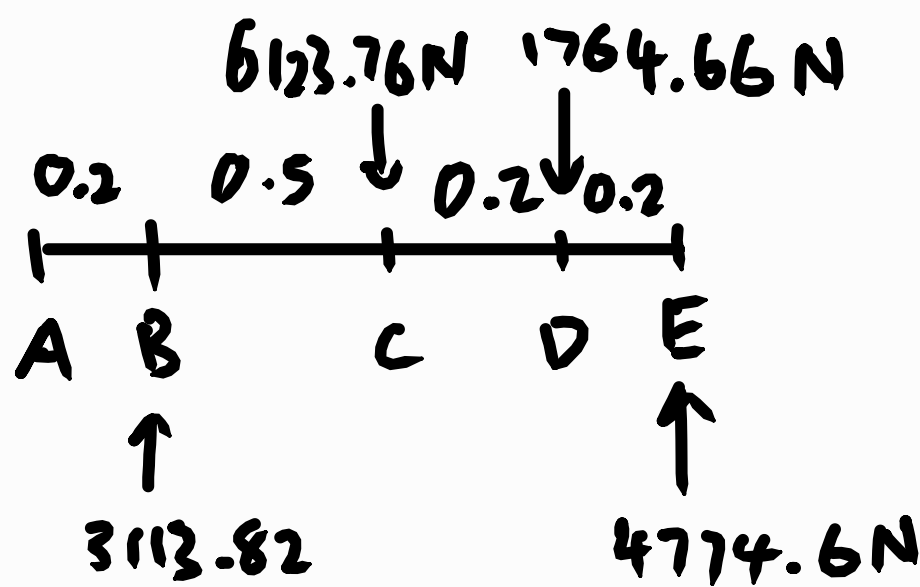
$$\sum F_y = 0$$

$$-1764.66 + 3113.82 + F_{C_y} + 4774.6 = 0$$

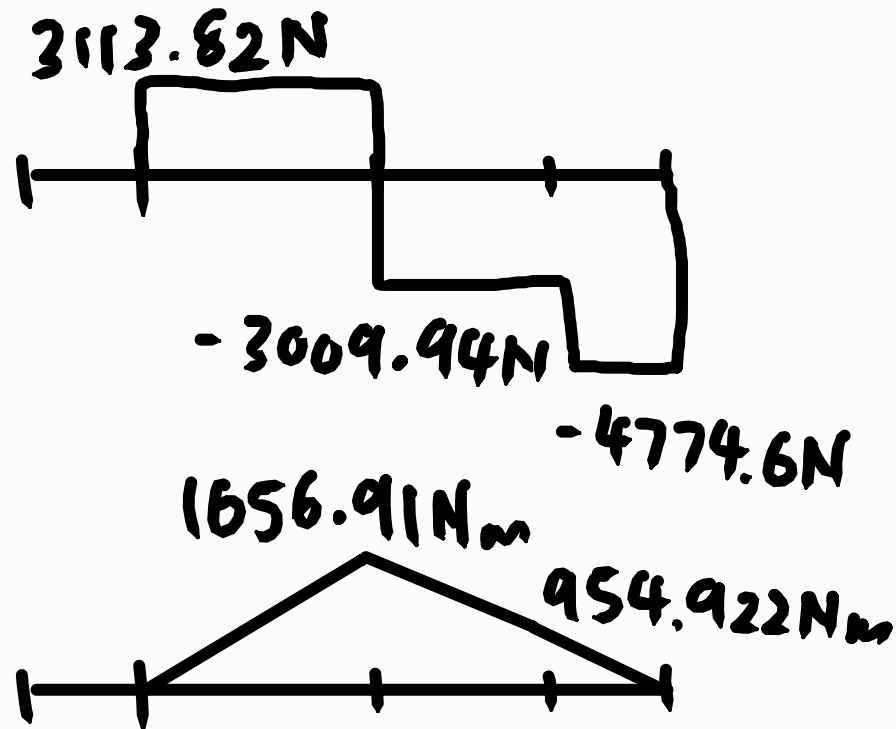
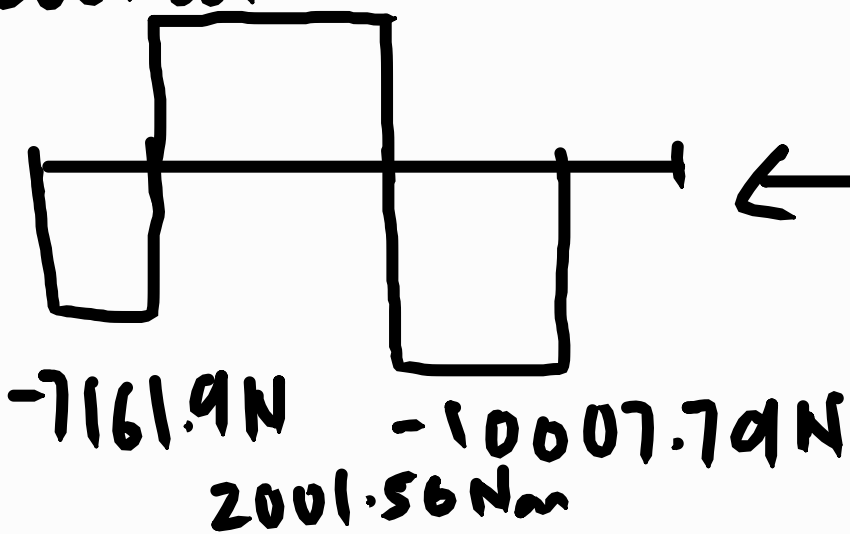
$$F_{C_y} = -6123.76 \text{ N } \downarrow$$



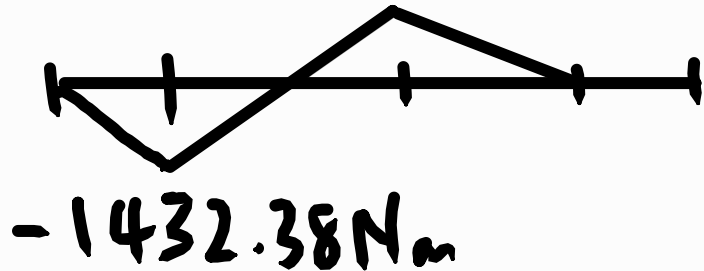
← Loading →



← Shear force →



← Bending moment →



Resultant moment:

$$M_A = 0$$

$$M_B = \sqrt{1432.38^2 + 0^2} = 1432.38 \text{ Nm}$$

$$M_C = \sqrt{2001.56^2 + 1656.91^2} = 2598.38 \text{ Nm}$$

$$M_D = 954.92 \text{ Nm}$$

$$M_E = 0$$

From appendix 3:

$$S_u = 676 \text{ MPa}$$

$$S_y = 565 \text{ MPa}$$

From figure 5-8:

$$S_n = 260 \text{ MPa}$$

From table 6-1:

$$C_R = 0.81$$

$$S_n' = C_R C_S S_n$$

$$= 0.81 \times 0.82 \times 260$$

$$= 172.69 \text{ MPa}$$



Shaft design equation:

$$D = \left[ \frac{32N}{\pi} \sqrt{\left( \frac{K_{fb} M}{s_n'} \right)^2 + \frac{3}{4} \left( \frac{T}{s_y} \right)^2} \right]^{\frac{1}{3}}$$

| Component     | Location | Feature        | $K_{fb}$ | $M(Nm)$ | $T(Nm)$ | $D(mm)$ |
|---------------|----------|----------------|----------|---------|---------|---------|
| A - Belt      | Left     | Retaining ring | 3        | 0       | 0       | 0       |
|               | Centre   | Press fit      | 1        | 0       | 477.76  | 28.17   |
|               | Right    | Well-rounded   | 1.5      | 0       | 477.76  | 28.17   |
| B - Bearing 1 | Left     | Well-rounded   | 1.5      | 1432.4  | 477.76  | 72.49   |
|               | Centre   | Press fit      | 1        | 1432.4  | 477.76  | 63.37   |
|               | Right    | Sharp fillet   | 2.5      | 1432.4  | 477.76  | 85.91   |
| C - Bearing 2 | Left     | Sharp fillet   | 2.5      | 2535.79 | 477.76  | 103.91  |
|               | Centre   | Press fit      | 1.0      | 2535.79 | 477.76  | 76.59   |
|               | Right    | Well-rounded   | 1.5      | 2535.79 | 477.76  | 87.65   |
| D - Gear      | Left     | Well-rounded   | 1.5      | 954.92  | 477.76  | 63.37   |
|               | Centre   | Profile key    | 2        | 954.92  | 477.76  | 69.71   |
|               | Right    | Retaining ring | 3        | 954.92  | 238.73  | 79.74   |
| E - Chain     | Left     | Well-rounded   | 1.5      | 0       | 238.73  | 22.36   |
|               | Centre   | Press fit      | 1        | 0       | 238.73  | 22.36   |
|               | Right    | Retaining ring | 3        | 0       | 0       | 0       |